

## Program 1:

The program “Density\_Comparison” measures the density of a compound with and without an element and compares them. It first searches the database for the density for the base elements, and then the elements that are added. It then converts each one into data frames. Then it plots the base elements on the left and the base elements + added to the right. Below is an example figure. The base elements I chose were “Fe” and “O”, and the added element I chose was “Al”. It then searched the database with just “Fe” and “O”, and “Al”, “Fe”, and “O”. When found, it extracted their densities and plotted them as you can see below.

```
if __name__ == "__main__":  
    # Change the elements in quotations to whatever is wanted.  
    analyze_element_effect(["Fe", "O"], "Al")
```

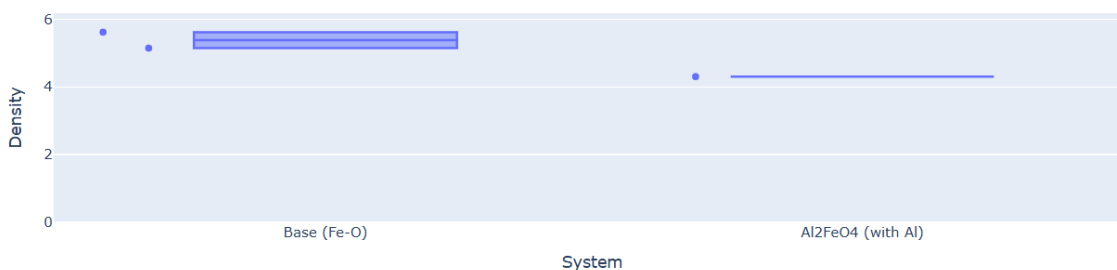
Analyzing how adding Al affects density of ['Fe', 'O'] compounds...

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Found 2 base compounds and 1 with added Al.

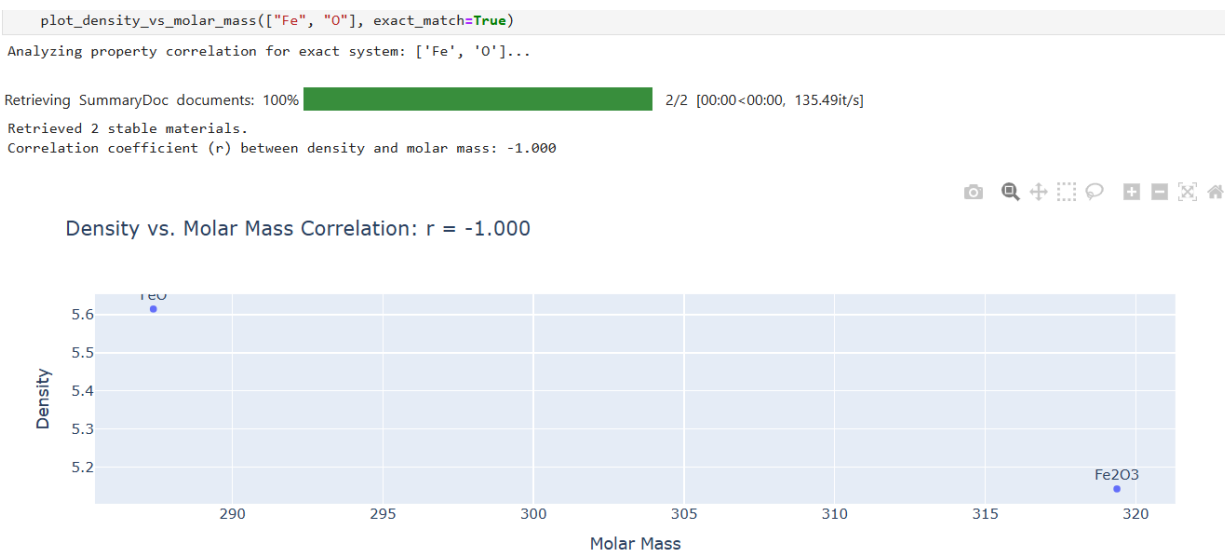
Effect of Adding Al on Density



## Program 2:

The program “Density\_vs\_Molarmass” compares the density and molar mass of specified elements. It first sees if the exact match is true, if it is, it will output only the compounds of the elements typed in. Otherwise, it will output compounds containing those elements. Then it extracts the density, molar mass, and formula of the found compounds/elements. It then creates a

data frame and correlates both properties. It then creates a scatter plot of density vs molar mass with the compounds plotted. The correlation value is listed in the title. Below is an example of “Fe” and “O” with exact match = True. As you can see, it only lists the compounds with those exact elements.



Below is an example of the same elements as before but with `exact match = False`. As you can see, it now lists compounds that contain “Fe” and “O”.

